

Data Card: Optimizing RAG Document Retrieval for Agronomic Advice

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Section	Guiding Questions	Your Response
Input vs. Output	• What are users inputting (for example, audio, text, images)? • What are the outputs or labels your model will predict or generate?	The retrieval model receives natural language text questions formulated by smallholder farmers or extension workers (e.g., describing visible crop symptoms, soil issues, or climate risks). The model outputs a ranked list of the top five most relevant document identifiers, evaluated using the nDCG@5 metric.
Labeling	• Where do your labels come from—domain experts, community members, crowd workers, AI-generated sources? • How will you check for consistency and accuracy?	Labels are provided through the predefined Kaggle competition dataset, featuring 4,194 query-document relevance judgments graded from 0 (irrelevant) to 3 (highly relevant). To ensure accuracy and reduce evaluation bias, agricultural experts will be considered in reviewing and assigning relevance judgments before real-world deployment.
Data creation	• How will you collect your data? • Will it come from direct sources (like interviews or recordings), or will you synthesize data in low-resource settings?	Data is entirely supplied through the competition corpus, comprising 695 agronomic documents, 308 training queries, and 200 test queries. The project does not involve independent data collection from direct sources, though the dataset does include some synthetic documents alongside verified, grounded extension literature.
Web collection ethics	• Do you plan to source data from the web? • If yes, which sites or platforms? • What tools will you use? • Have you addressed issues	We are strictly utilizing the provided corpus and do not engage in additional web scraping. Privacy is protected by not collecting personal data from farmers. Copyright is respected by utilizing appropriately

	of consent, copyright, and privacy?	licensed documents (e.g., CC-BY), and any future integration of real farmer queries will require informed opt-in consent.
Bias and representation	• Who might be left out of your dataset? • How will you ensure that the dataset reflects the diversity of the community and includes underrepresented voices?	Smallholder farmers lacking digital literacy, using colloquial dialects, or cultivating minor regional crops may be underrepresented due to a vocabulary mismatch with formal scientific documents. To improve fairness, we deploy a hybrid architecture combining lexical search with dense semantic models, ensuring queries describing visible symptoms still retrieve accurate guidance.
Values and community alignment	• How does your dataset design reflect the values you identified...? • How will your community benefit from this dataset, and how will you ensure accountability to them?	The design reflects values of accuracy, safety, fairness, transparency, and accountability by prioritizing exact matches to prevent crop-damaging advice. Communities benefit through dependable access to locally relevant evidence. We ensure accountability by returning verifiable source document identifiers and designing the system to augment human extension workers.